

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

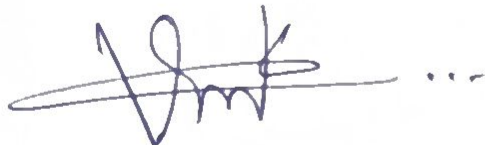
Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I V. Muralikrishna (192312587) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Co1, At 1

1, Scenario:- Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEFAOCAIVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

A, a) Given,

WECRLTEERDSOEFEFAOCAIVDEN
The system uses a 3-level transposition method.

Decryption.

Rail 1:- 1 5 9

Rail 2:- 2 4 6 8

Rail 3:- 3 7

The ciphertext is divided according to the no. of characters appearing on each rail.

Step 2:- Reconstruct the rails:-

Rail 1:- WECRLT

Rail 2:- EERDSOE

Rail 3:- FEOCAIVDEN

Step 3:- Read the characters following the Zig Zag pattern:-

plaintext:- WE ARE DISCOVERED FLEE AT ONCE

b, A transposition cipher only changes the positions of characters. It does not replace the characters themselves.

Major Weakness:-

1, Limited Key Space:-

A small no. of rails or arrangements can be tested quickly

2, Easily attacked using known plaintext:-

If an attacker recognizes a likely phrase or message format, the transposition is discovered

3, No authentication:-

It does not inherently verify whether the message has been modified by an attacker.

C, A practical improvement is to replace the legacy transposition cipher with a modern authenticated encryption algorithm such as AES-GCM.

plaintext



Transposition



Ciphertext

plaintext



AES-GCM Encryption



Ciphertext + Authentication tag



Secure transmission.

AES-GCM provides:- Confidentiality, Integrity.

Scenario:- Intercepted Suspicious Message:-

A cybersecurity intern intercepts the coded message:-

GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a, Deduce the most likely original message.

b, justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

a, The most likely cipher is the atbash cipher, which is a simple monoalphabetic substitution cipher.

In Atbash, the alphabet is reversed:

Now decrypt each word:-

Word 1,	Word 2,	Word 3,	Word 4,
G → T	R → I	Z → A	H → S G → T
S → H	H → S		V → E
R → I			X → C
H → S			I → R
			N → E

plaintext :- THIS IS A SECRET

b, The message can be identified as an atbash substitution cipher because each ciphertext character consistently corresponds to the character at the opposite end of the alphabet.

Example:-
G → T
S → H
R → I
H → S

This produces the meaningful word THIS

The preservation of spaces and word lengths is another indication of a simple substitution cipher.

- The one letter word Z is especially useful because in English a one-letter word is normally A or I. Under Atbash, Z maps directly to A.

Conclusion:-

Cipher text:- GSRH RH ZHXING

plain text:- THIS IS A SECRET

Although atbash provides simple confidentiality, it is not secure for modern communication because its substitution pattern is fixed and can be easily recognized and reversed.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation